A new sample preparation method for the determination of acyclovir by RP-HPLC: application to a drug-drug interaction study between gefitinib and acyclovir in rats

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Abstract: The assay of acyclovir in plasma seems to be a challenge because of its high hydrophily. In our present study, a reversed-phase high-performance liquid chromatography (RP-HPLC) method for the determination of acyclovir in rat plasma was described and validated in drug-drug interaction (DDI) between gefitinib and acyclovir in rats. The analytes were separated with gradient elution on C_{18} column (4.6 mm × 250 mm, 5 μm), and the peaks were recorded using ultraviolet detector at a wavelength of 254 nm. Protein precipitation followed by methyl tertiary butyl ether extraction was used for sample preparation. The calibration curve was established between 0.2 and 40 μg/mL (r^2 = 0.9999). The intra- and inter-day precisions were all less than 8%, and all the biases were not more than 10%. This new method was successfully applied to a DDI study between gefitinib and acyclovir in rats. Gefitinib up-regulated the absorption of acyclovir by about three times, and our findings guided the clinical co-administration of epidermal growth factor receptor-tyrosine kinase inhibitors (EGFR-TKIs) with acyclovir.

Keywords: Acyclovir; High-performance liquid chromatography; Sample preparation; Absorption; Drug-drug interaction

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1. Introduction

Acyclovir, a cyclic guanosine derivative, exhibits a selective inhibition of herpes virus replication (Fig. 1A)[1,2]. It is commonly used as a strong antiviral agent for the treatment of herpes simplex virus (HSV) infections in clinical practice[3]. Acyclovir is classified into the biopharmaceutics classification system (BCS) class 3 based on its characteristics of high solubility and low permeability. Due to the incomplete absorption, the bioavailability of acyclovir differs from 15% to 30%, and its pharmacokinetic parameters after oral administration are always highly variable[4,5].

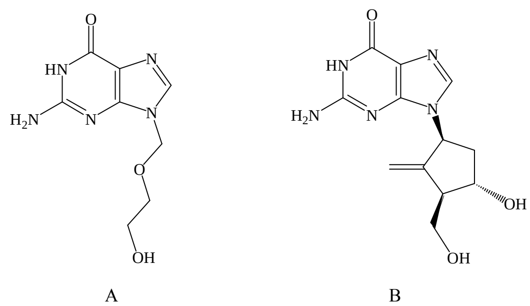

Figure 1. Chemical structures of (A) acyclovir and (B) entecavir.

Numerous methods have been applied in the assay of acyclovior in plasma, including radioimmunoassay[6], enzyme-linked immunoabsorbent assay (ELISA)[7], high-performance liquid chromatography method with fluorescence detection (HPLC-Flu)[8] and high-performance

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liquid chromatography method with ultraviolet absorbance (HPLC-UV)[9,10]. Radioimmunoassay and ELISA seem to be highly sensitive, but the pricing and supply of commercially available antiserum or monoclonal antibodies obstruct their applications, especially in larger samples. Regarding HPLC which is an economical and effective analysis tool for most drugs, acyclovir shows poor retention on the reversed-phase columns in the HPLC system owing to its high polarity[11]. Moreover, a similar structure to endogenous substances further complicates the analysis of acyclovir in plasma by HPLC method[12]. Although HPLC-Flu can provide a desired sensitivity and specificity, the high acidity of the mobile phase used to enhance the fluorescence response may significantly affect the lifetime of columns[13]. Furthermore, HPLC-UV methods with different sample preparation procedures have also been developed to determine the plasma levels of acyclovior. The protein precipitation by high concentrations of perchloric acid can be used for sample preparation of acyclovir in plasma, while the acid supernatant may lead to numerous late-eluting peaks and a significant reduction of the lifetime of the analytical column as well[9,10]. Solid-phase extraction yields good results, but this preparation is time- and cost-consuming[14]. Bahrami's group has published an HPLC-UV method with liquid-liquid extraction to analyze the acyclovir in plasma. Unfortunately, this assay includes a complicated mobile phase with a high flow rate[12]. Due to the above-mentioned reasons, it is highly necessary to develop an appropriate assay with good sensitivity, high selectivity, and effective sample preparation procedure.

Epidermal growth factor receptors (EGFRs) are major targets for anticancer therapy. Tyrosine kinase inhibitors (TKIs) are very effective against non-small cell lung cancer (NSCLC), which are presently recommend as first-line therapy of lung tumors caused by EGFR mutation[15]. Gefitinib is an orally active low-molecular-weight tyrosine kinase inhibitor that is highly selective for EGFR tyrosine kinase. It is the first EGFR-TKI and widely prescribed in NSCLC treatment since 2003[16,17]. Many cancer patients develop virus infection because of their hypoimmunity state resulting from the disease and chemotherapy agents. Since infection is a common problem in patients with cancer, the risk of drug-drug interaction (DDI) between gefitinib and co-administered anti-infective drug acyclovir has to be taken into consideration[18]. Up to now, no data on the effect of taking acyclovir concomitantly with gefitinib have been available.

Herein, we developed a simple HPLC-UV method with the novel preparation procedures to estimate acyclovir in plasma and applied the proposed method successfully to a DDI study between gefitinib and acyclovir in rats.

2. Materials and methods

2.1. Reagents and chemicals

Reference standards of acyclovir (purity 99.4%) and entecavir (Fig. 1B) (purity 99.7%, internal standard) were obtained from National Institutes for Food and Drug Control (Beijing, China). Gefitinib was supplied by ANGE Pharmaceutical Co., Ltd. (purity 99.8%, Nanjing, China). Acyclovir tablets (50 mg per tablet) used in the pharmacokinetic study were purchased from Kunming YuanRui Pharmaceutical Co., Ltd. (Kunming, China). Acetonitrile was of HPLC grade (Tedia, Fairfield, USA), and ammonium acetate (analytical grade) was provided by Guanghua Chemical Co., Ltd. (Guangzhou, China). Methyl tertiary butyl ether (MTBE, analytical grade) was obtained from Sinopharm Chemcial Reagent Co., Ltd. (Shanghai, China). Deionized water was purified in a Purelab classic system ELGA Labwater (Shanghai, China). All other reagents and chemicals were of analytical grade from commercial sources. Blank plasma was obtained from the rats supplied by Shanghai SLAC Laboratory Animal Co., Ltd. (Shanghai, China).

2.2. Instrumentation and conditions

The analysis was performed on an Agilent 1200 (Agilent Technology, USA). The system consisted of a degasser, quat pump, autosampler, thermostatted column compartment, and UV detector. The analysis was performed on an ODS column (4.6 mm × 250 mm, 5 μm, Dikma, China), and a Gemini C_{18} column (4 mm × 3.0 mm, 5 μm, Phenomenex, Torrance, USA) was employed as a guard column. The mobile phase for gradient elution consisted of two solvent systems: solvent A (acetonitrile) and solvent B (10 mmol/L ammonium acetate buffer). Gradient elution was carried out at a flow rate of 1 mL/min as follows: 97% (v/v) solvent B was used in the first 6 min, then the percentage of solvent B was decreased to 90% (v/v) until 10.50 min. Solvent B was retained at 10% (v/v) during 10.60–12.50 min, and it was up to 97% (v/v) during 12.60–15.00 min. The eluents were detected by a UV detector at a wavelength of 254 nm with an injection volume of 20 μL. All the analyses were performed at 30 ºC.

2.3. Preparation of stock and standard solutions

The stock solutions of acyclovir (400 μg/mL) and IS (1 mg/mL) were newly prepared with methanol and purified water (3:2, v/v) and stored at 4 ºC. The IS working solution was diluted to 50 μg/mL with methanol and purified water (3:2, v/v) before use. The standard solutions of acyclovir were prepared by diluting the stock solution with methanol and purified water (3:2, v/v) to 400, 200, 100, 50, 20, 10, 5 and 2 μg/mL. The calibration standards and quality control (QC) samples were prepared by adding 10 μL of standard solutions of acyclovir into 90 μL blank rat plasma. Finally, the concentrations of the calibration standards were 40, 20, 10, 5, 2, 1, 0.5 and 0.2 μg/mL. QC samples were independently prepared in the same way at three concentration levels (32, 5 and 0.5 μg/mL). All samples were freshly prepared daily.

2.4. Sample preparation

Briefly, 10 μL IS working solution (50 μg/mL) was added to 100 μL standard plasma. After vortex-mixing for 30 s, 200 μL acetonitrile was added. The mixture was then vortex-mixed for 3 min and centrifuged at 15 700 × g for 10 min. Then, 250 μL of the supernatant was transferred to a clean tube containing 50 μL 5% (v/v) formic acid solution. Subsequently, 1 mL MTBE was added into the tube next. After vortex-mixing for 3 min, the sample was centrifuged at 440 × g for 5 min, and the subnatant was evaporated to dryness under a gentle stream of nitrogen at 40 ºC. After dissolving the residue by 100 μL mobile phase, 20 μL liquid was injected into the HPLC system.

2.5. Method validation

2.5.1. Specificity

Specificity was assessed by comparing the chromatograms of extracted blank rat plasma from six different rats with the corresponding plasma samples spiked with acyclovir (5 μg/mL) and IS, and chromatograms of typical plasma samples from different rats after oral administration of 300 mg/kg acyclovir suspension for 4 h were also analyzed to check for the endogenous interference.

2.5.2. Linearity and lower limit of quantification (LLOQ)

The calibration curve for acyclovir was assessed by analyzing eight standard samples over the range of

0.2−40 μg/mL in plasma. The peak area ratio of acyclovir to the internal standard was plotted against acyclovir respective standard concentration. The regression parameters were calculated by linear square regression ($1/C^2$ weighing). The acceptance criterion for each back-calculated standard concentration was ±15% deviation from the nominal value, except for LLOQ (±20%). The LLOQ was the lowest concentration on the calibration curves of acyclovir measured with acceptable precision and accuracy. Six replicates of 0.2 μg/mL plasma samples were analyzed, and the concentrations of the samples were determined using the calibration curve.

2.5.3. Precision and accuracy

Precision and accuracy were evaluated by analyzing QC samples at three concentration levels (0.5, 5 and 32 μg/mL) on five replicates of each level during three continuous validation days. The concentration of each sample was detected using the calibration curve prepared on the same day. Precision was expressed in the relative standard deviation (RSD%). Accuracy was defined as the percentage of measured concentration to nominal concentration. The acceptance criterion was within ±15% bias of the nominal value and with a relative standard deviation of less than 15%.

2.5.4. Recovery

The recoveries of acyclovir were determined at three QC concentration levels (0.5, 5 and 32 μg/mL) on five replicates of each level and expressed as the percentage of the peak areas that were obtained from plasma samples to standard solutions without extraction procedure at the same theoretical concentrations. The recovery of the IS was evaluated at 5 μg/mL.

2.5.5. Stability

The stability was evaluated by analyzing QC samples at three concentration levels (0.5, 5 and 32 μg/mL) under the different conditions: at room temperature for 2 h and 5 h; before sample preparation (room temperature stability), and at sample injection temperature for 24 h after sample preparation (post-preparative stability). Freeze-thaw stability (three freeze/thaw cycles from –20 ºC to room temperature), long-term stability (–20 ºC for 60 d), and stock solution stability of acyclovir and IS at 4 ºC for 1 month have been reported by G. Holkar[19], G. Bahrami[12] and F. J. Zhao[20].

2.6. Pharmacokinetic study

2.6.1. Compliance with Ethical Standards

The validated method was applied to a pharmacokinetic study involving 12 Sprague Dawley (SD) rats, which were supplied by Shanghai SLAC Laboratory Animal Co., Ltd. (Shanghai, China). The experiment met the requirement of the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Before study initiation, the protocol of this study was approved by the Animal Ethics Committee of the Third Xiangya Hospital of Central South University (Changsha, China).

2.6.2. Sample collection

A total of 12 rats with six males and six females (body weight, (200 ± 20) g) were acclimatized to the experimental environment for 1 week before the study. The ranges of the temperature and humidity monitored during the study were (24 ± 1) ºC and 54% ± 3%, respectively. Rats were randomly allocated to two groups, with six rats in each group: the acyclovir control group (saline for 7 d plus 300 mg/kg acyclovir at the last administration) and the acyclovir experiment group (25.7 mg/kg/d gefitinib for 7 d plus 300 mg/kg acyclovir at the last administration). Next, 0.3 mL blood was collected into heparinized polypropylene tubes at 0 (pre-dose), 0.17, 0.33, 0.5, 0.75, 1, 1.5, 2, 3, 5, 7 and 9 h after administration. The sample was centrifuged immediately at 980 × g for 10 min at 4 ºC, and the plasma obtained was frozen at –20 ºC until analysis.

2.6.3. Pharmacokinetic analysis

The pharmacokinetic parameters of acyclovir, including maximum plasma concentration (C_{max}), the time point of maximum plasma concentration (T_{max}), elimination half-life ($t_{1/2}$), area under the plasma concentration versus time curve from 0 h to the last measurable concentration ($AUC_{0→t}$), area under the plasma concentration versus time curve from 0 h to infinity ($AUC_{0→∞}$), mean residence time from 0 h to the last measurable concentration ($MRT_{0→t}$), mean residence time from 0 h to infinity ($MRT_{0→∞}$), clearance (Cl/F) and apparent volume of distribution (Vz/F), were calculated by Phoenix WinNonlin 6.0 software (Pharsight Corporation, St. Louis, MO). Data were expressed as means ± SD, and the statistics software SPSS 11.5 (IBM, Armonk.NY) was used to analyze the variance. Statistical probability (P) in tables was expressed as $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$.

3. Results

3.1. Method development

Compared with previous reports[10,12–14], a simple mobile phase consisting of acetonitrile and 10 mmol/L ammonium acetate buffer with a gradient elution in our study could produce a good peak separation with appropriate retention time. To eliminate endogenous interferences from target peaks, 50 μL of 5% (v/v) formic acid was added to the supernatant after protein precipitation. Then the mixture was extracted with 1 mL MTBE, and the supernatant containing acyclovir and IS was evaporated to dryness. The typical chromatograms demonstrated that few endogenous substances interfered with acyclovir and IS in plasma.

3.2. Specificity

Figure 2 shows the typical chromatograms of blank plasma, blank plasma spiked with acyclovir (5 μg/mL), and rat plasma samples at 4 h after oral administration of 300 mg/kg acyclovir suspension. The retention time of acyclovir and IS was 9.4 and 12.4 min, respectively. No obvious interferences from endogenous substances were observed, indicating good specificity and selectivity.

3.3. Linearity and LLOQ

The calibration curve for acyclovir was linear over a concentration range of 0.2–40 μg/mL in plasma. The mean regression equation of the calibration curve was $Y = 0.2130X + 0.02207$ ($r^2 = 0.9999$). The LLOQ for acyclovir in plasma was 0.2 μg/mL, and the accuracy and precision of LLOQ were 96.3% and 6.1%, respectively.

3.4. Precision and accuracy

Table 1 shows the intra- and inter-day precision and accuracy values. As shown in the table, the intra-day and inter-day precision ranged from 0.2% to 0.9% and from 5.4% to 7.4%, respectively. The accuracy of the method ranged from 90.0% to 101.5% (intra-day) and from 95.5% to 102.0% (inter-day). The data demonstrated that the precision and accuracy of this assay were within the acceptable range.

3.5. Recovery

The recoveries of acyclovir from rat plasma were 95.5% ± 2.1%, 84.5% ± 5.4% and 82.9% ± 3.7% at the concentrations of 0.5, 5 and 32 μg/mL, respectively. The mean recovery of IS was 80.8% ± 5.9%, showing a satisfactory extraction recovery.

3.6. Stability

The results of stability shown in Table 2 indicated that acyclovir was stable in rat plasma and processed samples under the conditions described above.

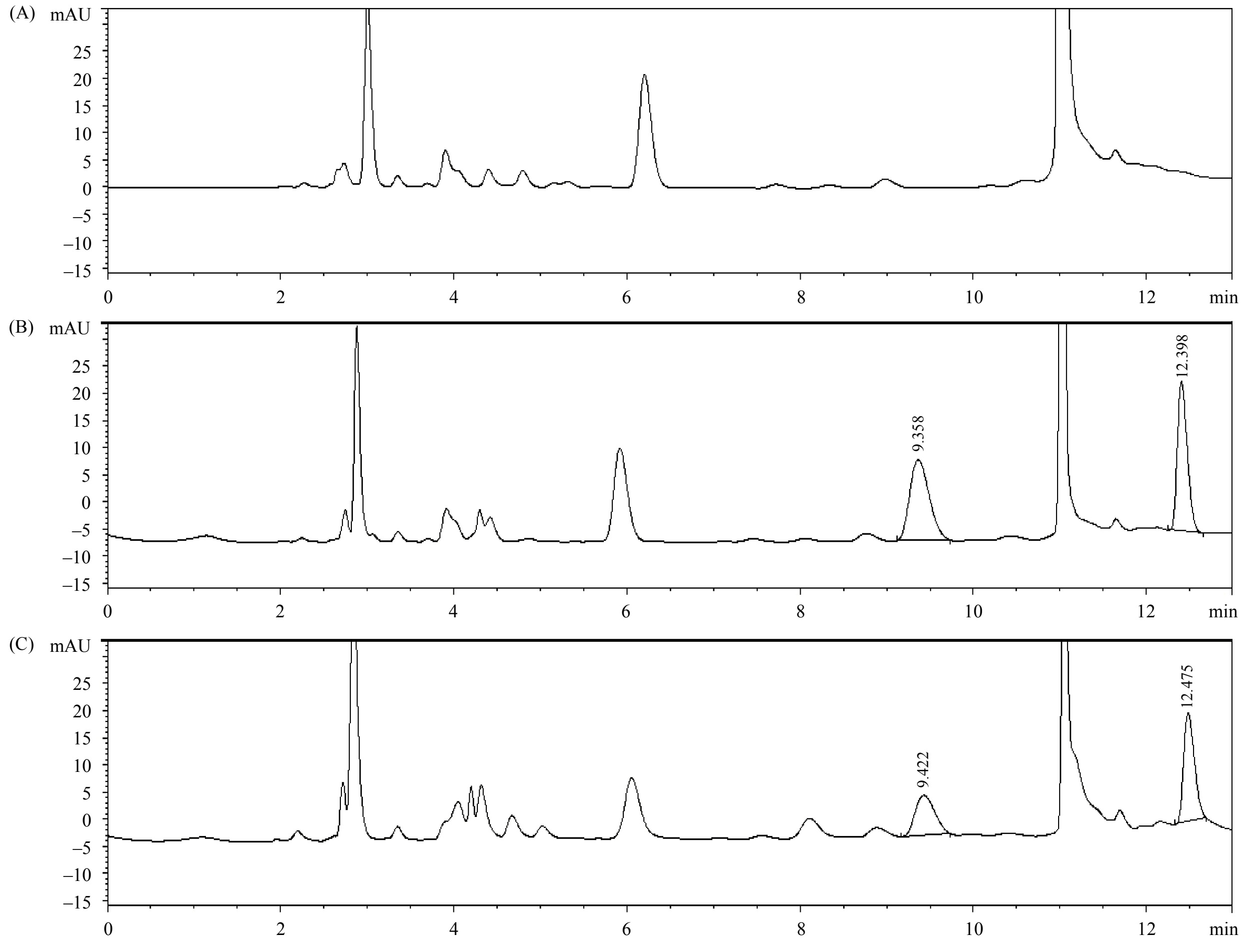

Figure 2. Typical chromatograms of acyclovir in rat plasma: (A) Blank rat plasma; (B) Blank plasma spiked with acyclovir (5 μg/mL) and IS; (C) Plasma sample from rat 4 h after oral administration of 300 mg/kg acyclovir suspension.

Table 1. Intra- and inter-day accuracy and precision of acyclovir in rat plasma.

Concentration added (μg/mL)	Intra-day (n = 5)			Inter-day (n = 3)		
	Mean concentration found (μg/mL)	Accuracy (%)	Precision (RSD%)	Mean concentration found (μg/mL)	Accuracy (%)	Precision (RSD%)
0.5	0.5077	101.5	0.9	0.5099	102.0	7.4
5	4.499	90.0	0.2	4.777	95.5	5.4
32	29.69	92.8	0.3	31.21	97.5	5.4

Table 2. Stability of acyclovir in rat plasma under various conditions (n = 3).

Condition	0.5 μg/mL			5 μg/mL			32 μg/mL		
	Mean (μg/mL)	Accuracy (%)	RSD (%)	Mean (μg/mL)	Accuracy (%)	RSD (%)	Mean (μg/mL)	Accuracy (%)	RSD (%)
Post-preparative stability (24 h)	0.5131	102.6	0.9	4.53	90.6	0.2	30.01	93.4	0.4
Room temperature stability (2 h)	0.5043	100.9	0.5	4.956	99.1	2.6	31.34	98.0	2.8
Room temperature stability (5 h)	0.5155	103.1	2.9	5.169	103.4	1.2	34.04	106.4	1.1

Table 3. The pharmacokinetic parameters of acyclovir with and without the co-administration of gefitinib (n = 6).

Parameters (unit)	With gefitinib	Without gefitinib
$t_{1/2}$ (h)	3.1 ± 1.4	2.4 ± 0.7
C_{max} (μg/mL)	6.8 ± 1.9**	2.6 ± 1.1
T_{max} (h)	1.5 ± 0.5	1.3 ± 0.5
$AUC_{0\rightarrow t}$ (μg/mL)	33.6 ± 3.9***	10.7 ± 4.4
$AUC_{0\rightarrow\infty}$ (μg/mL)	39.6 ± 5.7***	12.0 ± 5.2
$MRT_{0\rightarrow t}$ (h)	3.8 ± 0.2***	3.1 ± 0.3
$MRT_{0\rightarrow\infty}$ (h)	5.4 ± 1.5*	4.0 ± 0.6
Vz/F (L/kg)	35.9 ± 21.2**	100.3 ± 46.3
CL/F (L/kg/h)	7.7 ± 1.2***	29.8 ± 13.2

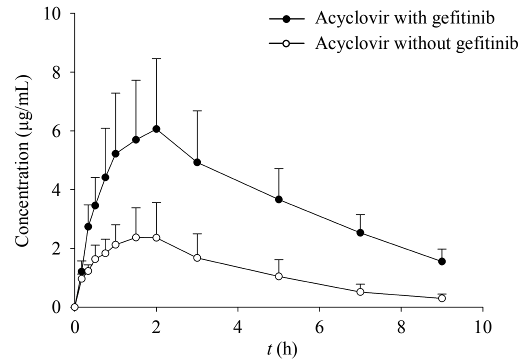

Figure 3. Gefitinib increased the absorption of acyclovir in rats. Error bars represented the standard error of the mean (n = 6).

3.7. Pharmacokinetic study

The proposed analytical method was successfully applied to a DDI study between gefitinib and SD rats. Table 3 lists the pharmacokinetic parameters of acyclovir in rats, and the average concentration-time profiles were also exhibited in Figure 3. Compared with the results from the experiment that acyclovir was administered alone, the C_{max} of acyclovir was increased from 2.6 to 6.8 μg/mL, and $AUC_{0\rightarrow t}$ was also increased from 10.7 to 33.6 μg/mL after acyclovir was co-administered with gefitinib. The results showed a DDI between acyclovir and gefitinib that gefitinib could significantly increase the absorption of acyclovir when they were used together.

4. Discussion

Acyclovir is a small molecule hydrophilic anti-virus drug. As a low-mass compound with high polarity, acyclovir shows poor retention on conventional reversed-phase columns. Moreover, it is structurally similar to endogenous substances, so the analysis of acyclovir in plasma is very complicated. Different from traditional analytical methods, we detected acyclovir without using expensive solid-phase extraction. A strong acidic mobile phase and a high mobile phase flow rate, which would decrease the lifetime of column and instrument, could be avoided at the same time. In the present work, an economical and effective HPLC-UV method with a simple sample preparation process was built for the determination of acyclovir in rat plasma samples. Moreover, it provided a new idea for evaluating plasma levels of water-soluble drugs, such as peramivir, entecavir, adefovir, and tenofovir.

Nowadays, the incidence of malignant tumors is increasing in clinical practice, which seriously affects the life quality and even safety of patients. Suffering from malignant tumors, the body function and resistance of patients have been greatly reduced. Moreover, after long-term radiotherapy and chemotherapy, there will be different degrees of fatigue, myelosuppression, loss of appetite, and other adverse reactions, which further reduce the immunity of patients[21,22]. Clinically, herpes zoster frequently occurs as a common skin disease, especially for people with low immunity. EGFR-TKIs and co-administered anti-infective drug acyclovir are recommended for NSCLC patients with herpes zoster[23]. Simotinib (SIM6802) is one of the new orally active EGFR-TKI anticancer drugs currently under clinical trials. It is prescribed for patients with NSCLC like gefitinib. The results of our previous study have demonstrated that simotinib increases the exposure of acyclovir in combination, suggesting a DDI between EGFR-TKIs and acyclovir. As a representative drug of EGFR-TKI, gefitinib is commonly used in combination with acyclovir to control viral infections in NSCLC patients who suffer from hypoimmunity[24]. The results of this study revealed that gefitinib might upregulate the absorption of acyclovir by increasing the $C_{\max}$ and $AUC_{0\rightarrow t}$ by about three times. The results were consistent with those observed from the DDI study between simotinib and acyclovir. The probable mechanism of the DDI was that EGFR-TKIs could reduce the expression of the cell junction gene afadin-6 and increase the paracellular permeability of intestinal epithelial cells, which led to the increase of acyclovir exposure[25]. The results suggested that more attention might be paid to the dose adjustment of acyclovir when it was used in combination with EGFR-TKIs like simotinib or gefitinib.

In conclusion, we developed an economical, simple, and reproducible RP-HPLC method with a new sample preparation procedure for the determination of acyclovir in rat plasma. The proposed method was successfully applied to a DDI study between gefitinib and acyclovir in rats. It provided a new idea for evaluating plasma levels of water-soluble drugs, such as peramivir, entecavir, adefovir, and tenofovir. Importantly, the interesting finding of this work reminded us that there might be a propensity for DDI when acyclovir was concomitantly used with EGFR-TKIs like gefitinib.

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高效液相色谱法测定大鼠血浆中阿昔洛韦浓度及其与吉非替尼药物相互作用的研究

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摘要: 由于阿昔洛韦水溶性高, 较难用普通高效液相色谱法测定其血浆中的药物浓度。本研究建立并验证了测定大鼠血浆中阿昔洛韦浓度的高效液相色谱方法, 并将该方法应用于与吉非替尼的药物相互作用研究中。采用Platisil ODS柱(4.6 mm × 250 mm, 5 μm, 迪马科技), 检测波长254 nm, 蛋白沉淀后再用甲基叔丁基醚萃取进行样本处理。阿昔洛韦浓度在0.2–40 μg/mL范围内具有良好的线性关系(r^2 = 0.9999), 本方法批间和批内精密度均小于8%, 批间和批内准确度偏差不超过10%。该方法已成功应用于大鼠体内吉非替尼和阿昔洛韦的药物相互作用研究中。结果显示, 同时服用吉非替尼可使阿昔洛韦的吸收增加3倍, 这为临床表皮生长因子-酪氨酸酶抑制剂与阿昔洛韦的合用提供了参考。

关键词: 阿昔洛韦; 高效液相色谱法; 样本处理; 吸收; 药物相互作用